

REPORT ON MPPL ABU ROAD MP1 FUEL PUMP CAM EYE FAILURE

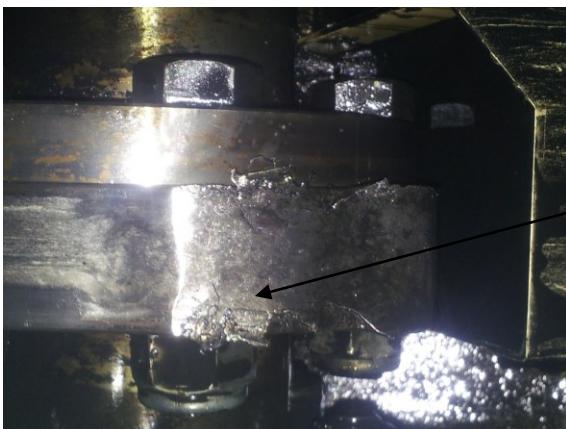
INTRODUCTION:

On 08.08.2012, MPPL Abu Road MP1 (Type: 8S 37G HBC; Sr No D6/50161-6) was started on trial run for ascertaining the cause of yellow smoke and low temperature in cylinder 7. On increasing or decreasing the fuel supply via fuel rack in cylinder number 7, no change in engine sound was observed. Also there was a particular sound coming from cam door of cylinder 7. The cumulative running hours of engine as on date was 54412 hrs.

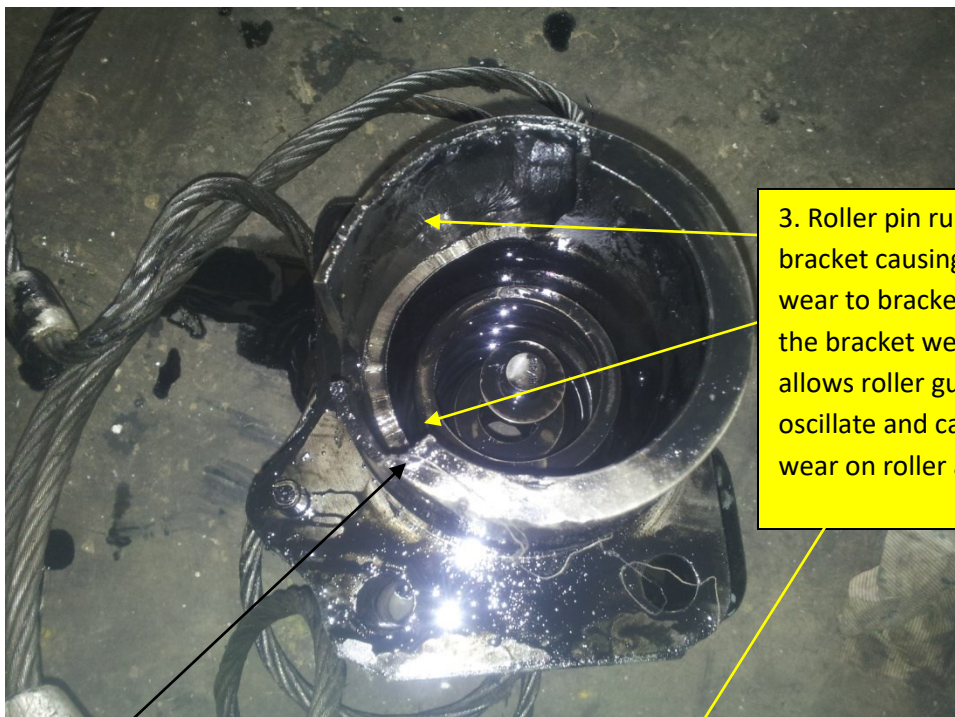
The last major overhaul (9000 hrs maintenance) of engine was done at 51806 hrs in Nov 2011 to Jan 2012. Cam eye has not been changed since the commissioning of the engine hence running hours of the component is same as the engine running hours.

SEQUENCE OF EVENTS

- 1) On 28 July 2012, yellow smoke was observed on starting the engine. Other parameters of the engine were found to be normal. Upon inspection it was found that water has mixed in the oil of oil bath filter. 3000 hrs maintenance was carried out and oil bath filter was cleaned and oil changed.
- 2) On 2 August 2012, engine was started on trial run but the problem persisted. There was leakage of gear box oil from NDE side deflector. Gear box leakage problem was rectified.
- 3) On 8 August 2012, engine was again started on trial run. Problem of low temperature in cylinder 7 was observed. At low rpm each cylinder was individually checked to confirm whether the problem was with fuel side. In cylinder 7, on fully opening or closing the fuel rack no change in engine sound was observed suggesting that firing was not taking place in cylinder 7. Also there was a peculiar sound coming from cam door of cylinder 7. Upon opening of cam door of cylinder 7, fuel pump bracket, roller guide and cam eye were found to be damaged.

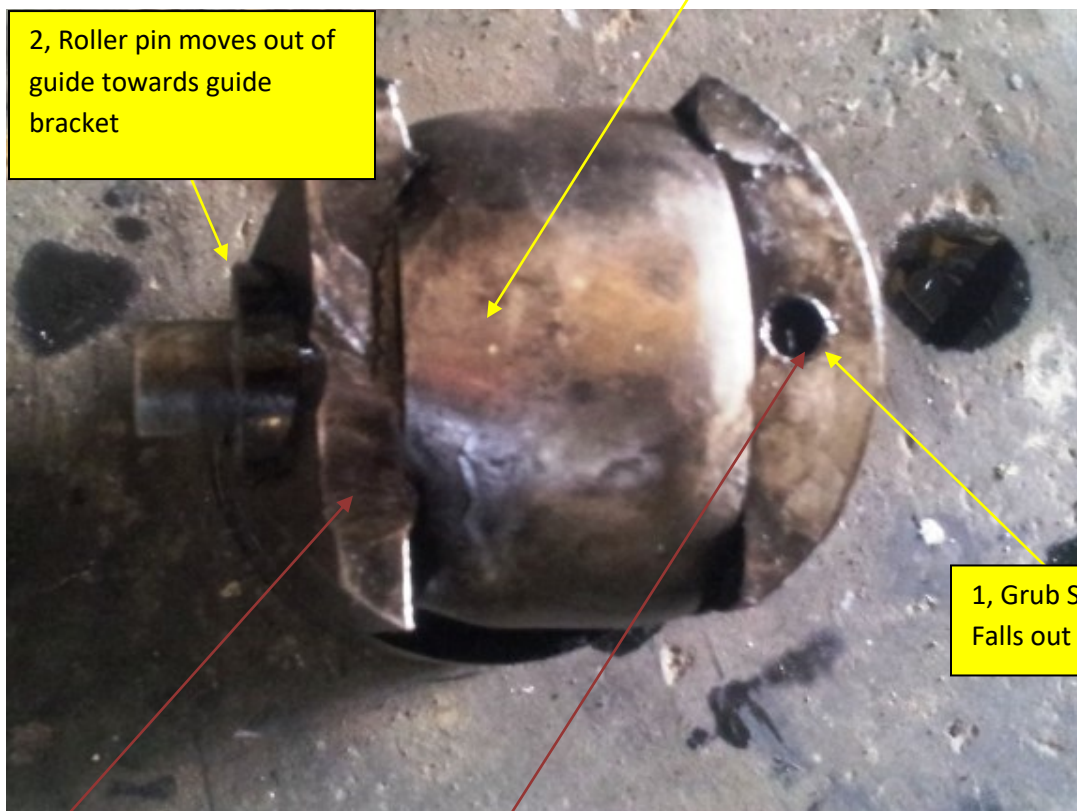


Damage in Fuel Pump
cam



3. Roller pin rubs on guide bracket causing excessive wear to bracket. As slot in the bracket wears this allows roller guide to oscillate and causes uneven wear on roller and fuel cam

Material broke from fuel pump bracket



2, Roller pin moves out of guide towards guide bracket

1, Grub Screw Falls out

Damaged Roller guide

Pin not found here

DETAILS OF FUEL PUMP CAM EYE

Part No: SD57924

001365/9

F6267

STATUS OF OTHER FUEL PUMP CAM EYE

All the remaining fuel pump cam eye in MP-1 were checked for defects. Only minor defect was found in fuel pump cam eye of cylinder number 8 which has been rectified. Remaining all were found to be ok.

PROBABLE CAUSE OF FAILURE:

- Roller guide came out from its slotted position in the fuel pump bracket due to material failure of fuel pump bracket. During running of the engine roller guide got twisted from its actual position. This caused shear load on the fuel pump cam eye elongating the material along the edges.

RESTORATION WORK

Two pair of old fuel pump cam eye were brought from Sanganer to be replaced in cylinder number 7 and cylinder number 8 of MP-1, MPPL, Abu Road. New fuel pump bracket and roller guide taken from Abu Road stores were fitted in cylinder 7.